

Total Synthesis of Hoiamide A Using an Evans–Tishchenko Reaction as a Key Step

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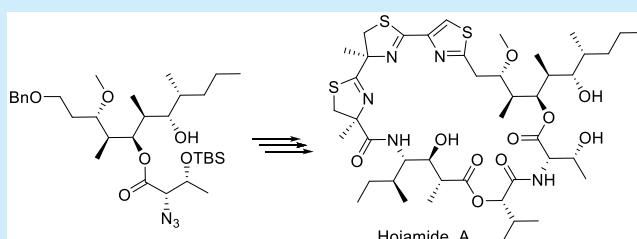
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Supporting Information

ABSTRACT: The first total synthesis of neurotoxic cyclopeptide hoiamide A (**1**) has been accomplished. The synthesis features the use of an Evans–Tishchenko fragment coupling between a five-stereogenic-center-containing β -hydroxyketone and a chiral aldehyde derived from threonine.



Hoiamides (Figure 1) are extraordinary cyclodepsipeptides, isolated from an environmental assemblage of the marine

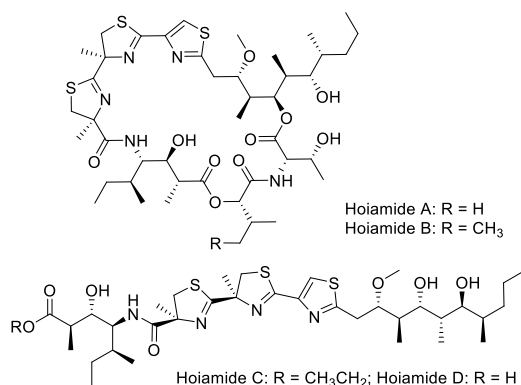


Figure 1. Structures of hoiamides A–D.

cyanobacteria *Lyngbya majuscula* and *Phormidium gracile* collected in Papua New Guinea.¹ Hoiamide A (**1**) contains 15 stereogenic centers embedded within a 26-membered macrolactone including a highly modified peptidic unit featuring an acetate extended and *S*-adenosyl methionine altered isoleucine residue, three consecutive five-membered heterocycles, and a C15-polyketide motif.^{1a} Characterization of the natural structure and determination of the relative and absolute stereochemistry of hoiamide A were conducted through a combination of integrated spectroscopic techniques and chemical degradation. It was suggested that the unique Dmetua (5,7-dihydroxy-3-methoxy-4,6,8-trimethylundecanoic acid) fragment may be derived biogenetically from propionates and an acetate. Preliminary studies suggested that hoiamide A acts as

a partial agonist at site 2 of VGSC α subunit and displays cytotoxicity toward a range of human cancer cell lines.^{1a} Further biological tests disclosed that hoiamide A can induce neuronal death via necrotic and apoptotic pathways at nanomolar concentrations.^{1d} The ability of hoiamide A to interact with some important therapeutic targets has allowed it to perform the important function as a probe molecule in chemical biology and also as a potential lead compound in the treatment of neurological disorders.^{1a,d}

As the first step in a program aiming at the biological evaluation of marine natural products and their analogues, we have undertaken the synthesis of several marine natural products,² which included the total synthesis hoiamide C,³ a linear analogue of hoiamide A. We describe in this Letter the development of an Evans–Tishchenko reaction-based strategy for assembling the complex ester fragment of hoiamide A, which culminated in the first total synthesis of this unique cyclodepsipeptide.

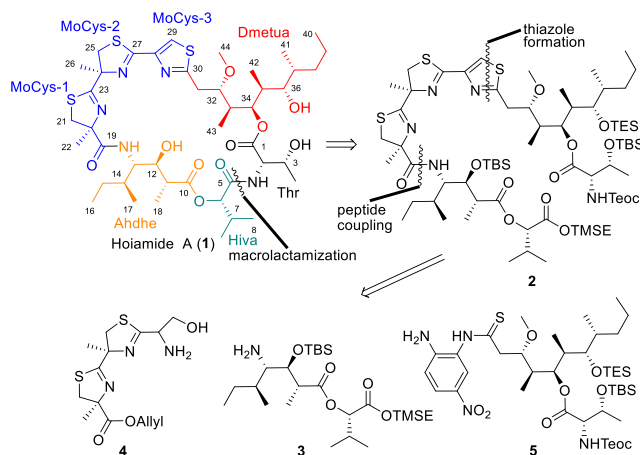
The synthetic challenges with regard to the synthesis of hoiamide A are the effective macrocyclization and the successful achievement of a complex ester fragment coupling in the synthesis of the polyketide subunit. As depicted retrosynthetically in Scheme 1, hoiamide A may be obtained from macrolactamization of precursor **2**, which was envisioned to arise from the assembly of three subunits **3**–**5**.

The synthesis of fragment **3** (Scheme 2) commenced with a 2-bromo-1-ethyl-pyridinium tetrafluoroborate (BEP)⁴ mediated condensation of the known carboxylic acid **6**³ with the known chiral hydroxy ester **7**,⁵ which provided the desired ester **8** in

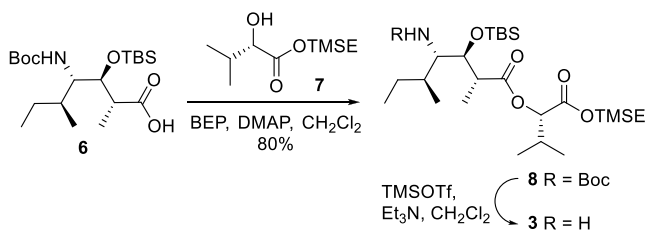
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Scheme 1. Retrosynthetic Analysis of Hoiamide A (1)



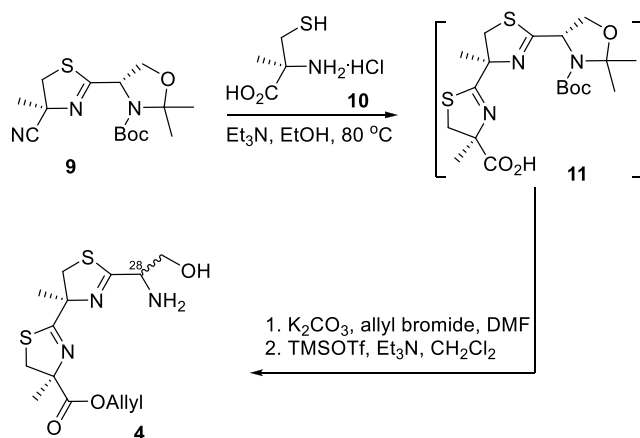
Scheme 2. Synthesis of Intermediate 3



80% yield. Removal of the Boc carbamate functionality of **8** was achieved using Ohfuné procedure⁶ to give rise to the corresponding free amine (**3**) that was sufficiently clean and could be used directly at a late stage of the synthesis (*vide infra*).

The synthesis of the bithiazoline-containing amino alcohol **4** started with the known nitrile **9**³ and (*S*)-2-methylcysteine hydrochloride (**10**), following the procedure previously described by Pattenden,⁷ which led to the thiazoline acid **11** (Scheme 3). Treatment of **11** with allyl bromide and potassium

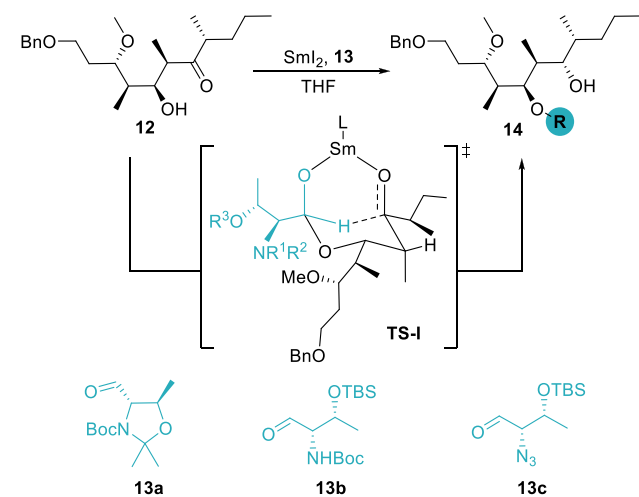
Scheme 3. Synthesis of Hydroxyl Amine 4



carbonate afforded the corresponding allyl ester; TMSOTf-triethylamine successfully cleaved both Boc and acetonide to provide hydroxyl amine **4** ready for further elaboration. In this event, the unexpected epimerization at C28 stereogenic center occurred. However, both epimers could be converted into the desired thiazole (*vide infra*).

With the modified isoleucine (**3**) and triheterocyclic fragment (**4**) in hand, we turned attention to the central fragment **5**. We envisaged that the combined severe steric hindrance around both the C34 hydroxyl and C1 carboxylic groups would pose significant challenges in the direct esterification with carboxylate activation. To circumvent this potential problem, an alternative strategy was considered. The 1,3-*anti*-diol monoester motif present in the polyketide substructure **5** presents an opportunity to use an Evans–Tishchenko coupling reaction.⁸ We rationalized that the fragment **5** could be constructed from β -hydroxyketone **12**, a key intermediate previously employed in our synthesis of hoiamide C.³ In this event, three threonine-derived aldehydes were employed as the coupling partners with β -hydroxyketone **12** in Evans–Tishchenko reaction. Depicted in TS-I, the process presumably involved an intramolecular hydride transfer from an intermediary hemiacetal, which was catalyzed by in situ formed $\text{SmI}_2 \cdot \text{SmI}(\text{RCHO})_2$.^{8b–f} As shown in Table 1, exposure of aldehyde **13a**⁹ and β -hydroxyketone **12** to

Table 1. Evans–Tishchenko Reactions of **12** with Aldehydes **13a–c**^a



entries	aldehyde	SmI_2 [equiv]	temperature [°C]	time [h]	yield [%]
1	13a	0.3	25	12	trace
2	13a	0.5	25	12	18
3	13a	0.8	25	12	22
4	13a	1.0	25	12	37
5	13b	1.0	25	12	43
6	13b	1.0	0	12	49
7	13b	1.0	−20	12	54
8	13b	1.0	−20	24	56
9	13b	1.0	−40	12	55
10	13c	1.0	25	12	47
11	13c	1.0	−10	12	56
12	13c	1.0	−20	12	86

^aYields of material isolated by silica gel chromatography. Diastereoselectivities (>95:5) were obtained by NMR analysis. See Supporting Information for further experimental details.

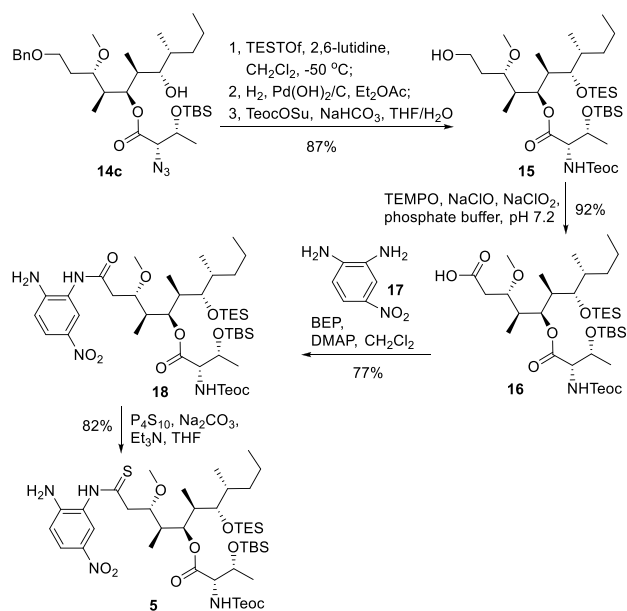
the standard reaction conditions (30 mol % of SmI_2 in THF at ambient temperature) gave complex product mixtures along with significant recovery of β -hydroxyketone **12**, but no desired product **14a** was isolated (entry 1). This observation led us to propose that more SmI_2 might be required to drive the reaction to completion. To our delight, increasing the quantity of SmI_2 from 0.3 to 1 equiv increased the yield up to 37% (entry 4).

When the less sterically crowded aldehyde **13b**¹⁰ was used in place of **13a**, the yield of the desired 1,3-*anti*-diol monoester **14b** increased to 43%.

Mechanistically we assume that the retro-aldol reaction and the formation of a transient samarium enolate might be responsible for the low yield. Fortunately, these side reactions could be largely suppressed at lower temperatures. In the event, lowering the reaction temperature to $-20\text{ }^{\circ}\text{C}$ improved the yield substantially (entry 7); however, extending the reaction time (entry 8) or further lowering the temperature to $-40\text{ }^{\circ}\text{C}$ (entry 9) had almost no noticeable effect on the outcome of the reaction. Searching for ways to reduce steric hindrance as much as possible, we decided to test whether the less sterically encumbered azido aldehyde **13c**¹¹ would be compatible with the Evans–Tishchenko coupling protocol. To our delight, Evans–Tishchenko coupling of β -hydroxyketone **12** with **13c** at $-20\text{ }^{\circ}\text{C}$ provided the monoesterified *anti*-diol **14c** in 86% yield as the sole product, and it is noteworthy that this reaction can be scaled effectively on a gram scale without compromising yield.

After the successful synthesis of **14c**, we now concentrated on its further elaboration (Scheme 4). Thus, a sequential functional

Scheme 4. Synthesis of Thioamide 5

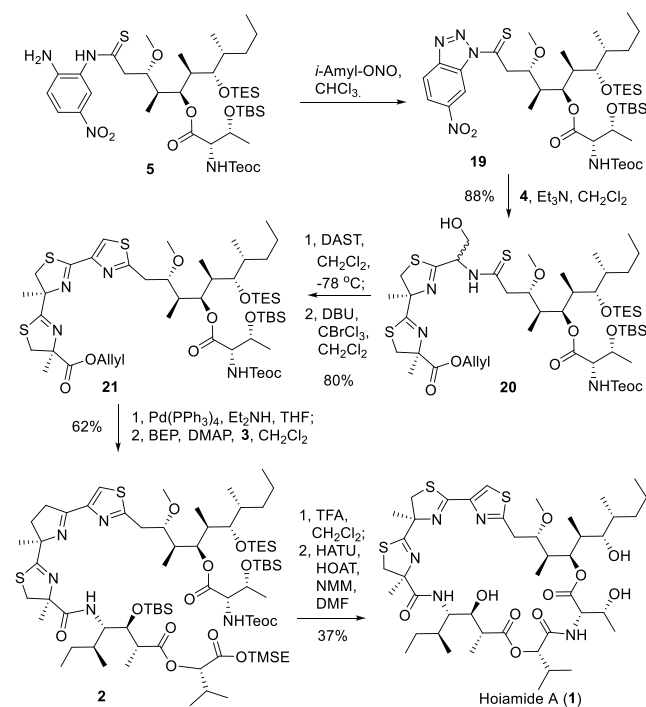


group transformation and protecting group manipulation of **14c** involved protection of the secondary hydroxyl group as its triethylsilyl ether by careful exposure to TESOTf in dichloromethane in the presence of 2,6-lutidine, concomitant reduction of the azido group, and removal of the benzyl ether under hydrogenative conditions, followed by protection of the resultant amine as its *N*-Teoc carbamate **15** in 87% yield over three steps. Oxidation of the primary hydroxy group of **15** by the sequential action of TEMPO/NaClO and NaClO₂ afforded the corresponding acid **16**,¹² which was then condensed with 4-nitro-1,2-phenylenediamine¹³ through the action of BEP/DMAP to give rise to **18** in 77% yield. Treatment of **18** with P₄S₁₀ and Na₂CO₃ in THF at ambient temperature provided the corresponding thioamide **5** in 82% yield in which the addition of triethylamine was required to prevent the undesired formation of desilylated products (Scheme 4).

With three fragments **3**, **4**, and **5** in hand, the completion of the synthesis of hoiamide A proved straightforward. In this

event, diazotization of thioamide **5** furnished the corresponding thionoacid derivative of nitrobenzotriazole (**19**),¹³ which reacted cleanly with the amino alcohol **4** to produce the hydroxy thioamide **20** in 88% yield (Scheme 5). Cyclo-

Scheme 5. Completion of Total Synthesis



dehydration of hydroxy thioamide **20** to the intermediate thiazoline was accomplished using diethylaminosulfur trifluoride (DAST)¹⁴ in CH₂Cl₂ at $-78\text{ }^{\circ}\text{C}$, followed by oxidation (DBU, BrCCl₃)¹⁵ of the resulting thiazoline to give rise to the thiazole **21** in 80% yield. Allyl ester cleavage was effected in THF containing dry diethylamine and a catalytic amount of tetrakis(triphenylphosphine)palladium to furnish the corresponding acid that was then condensed with intermediate **3** through the action of BEP/DMAP to afford macrocyclic precursor **2** in 62% yield over two steps. Simultaneous deprotection of the 2-(trimethylsilyl)ethyl (TMSE) ester, 2-trimethylsilylethyl carbamate (Teoc), and three silyl ether protecting groups by treatment with TFA afforded the desired amino acid, which was immediately activated by HATU/HOAT¹⁶ to afford hoiamide A (**1**) in 37% overall yield. The analytical data of hoiamide A, obtained from this total synthesis, were found to be in full agreement with those reported for hoiamide A isolated from natural source.

In summary, the execution of a highly convergent strategy has led to the completion of the first total synthesis of hoiamide A. The synthesis was accomplished in 15 steps over the longest linear sequence and in a 7% overall yield starting from readily accessible β -hydroxyketone **13**. The successful route hinged upon the early formation of the sterically encumbered ester bond by successful optimization of the Evans–Tishchenko reaction, the judicious choice of orthogonal protecting groups to facilitate selective manipulation of advanced intermediates, and the macrocyclic precursor. Furthermore, this synthesis should allow for the preparation of various analogues of hoiamide A for further exploration of their biological potential.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.orglett.9b01735](https://doi.org/10.1021/acs.orglett.9b01735).

Experimental details and data (PDF)

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Notes

The authors declare no competing financial interest.

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